# 🇩🇪 Germany — Monthly Intelligence Report v4.2

## Edition 006 — The Bavaria Solar Corridor

Where the energy transition is outrunning the grid. edition · August 2026 · SSI v4.2 (refreshed June 2026)

CONTENTS

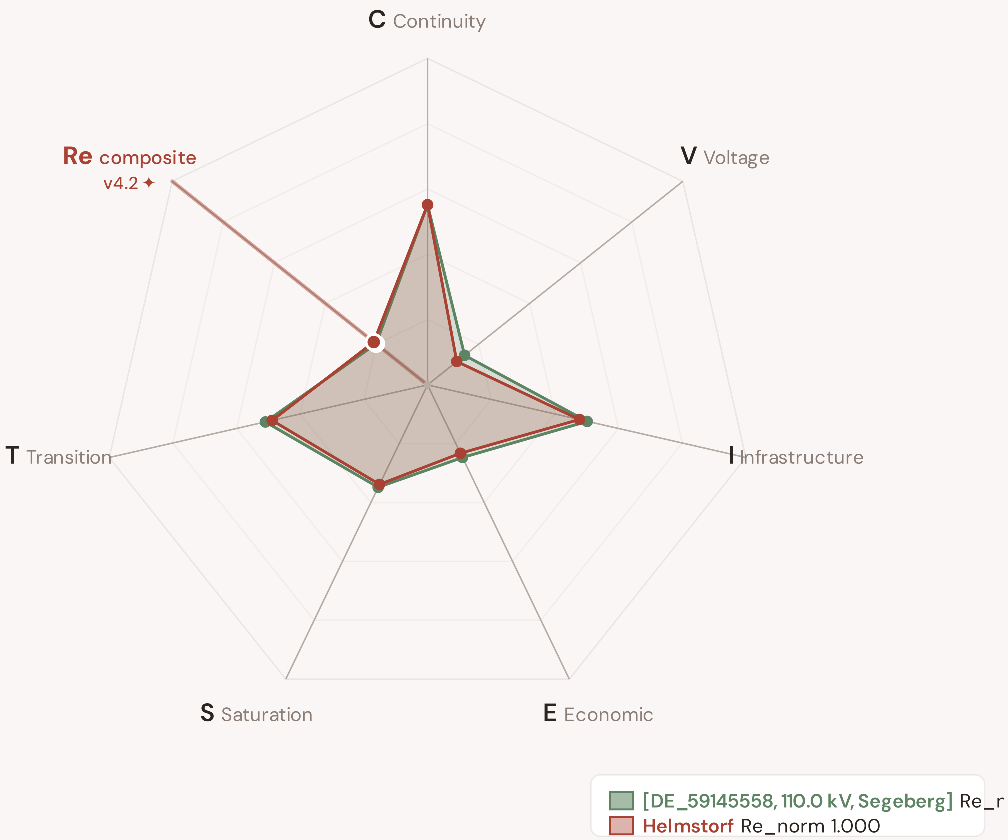


SECTION A

## Executive Summary

## Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw − 0.920) / (1.787 − 0.920), 0, 1)). Two worked examples overlaid: **[DE_59145558, 110.0 kV, Segeberg]** (low-exposure reference, Re_norm 0.000) · **Helmstorf** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.




*The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula `Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c − 1.00)`, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared `intelligence-sections.js` and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).*

SUBSTATIONS SCORED

**108,016**

4,462 EHV · 4,735 HV · 98,819 distribution-tier

GRID LINES MAPPED

**262,134**

~92,324 km coverage

MC SIMULATIONS

**42.9 M**

10,000 per substation

PUBLIC DATA SOURCES

**34**

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 108,016 substations and 262,134 grid lines spanning ~92,324 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

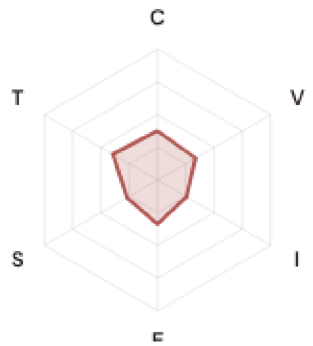
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Germany's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

### Substation in Focus

**Fürstenberg**

DE40A, Oberhavel · 110 kV

| 0.646 | High | 0.242 | 64th |
|---|---|---|---|
| R_FINAL | CLASSIFICATION | CI WIDTH | FLEET PERCENTILE |




**Re composite** v4.2 7th axis

**Re_raw** 1.4954 · **Re_norm** 0.664

ELEVATED EXPOSURE ⓘ

COMPONENTS

- **C** Continuity: 0.376 / 0.30 (125%)
- **I** Infrastructure: 0.257 / 0.25 (103%)
- **S** Saturation: 0.269 / 0.20 (135%)
- **V** Voltage: 0.337 / 0.10 (337%)
- **E** Economic: 0.337 / 0.10 (337%)
- **T** Transition: 0.394 / 0.05 (789%)

MODIFIERS

- **R3 Consequence:** 0.972 ↓ dampens
- **R4 Graph Criticality:** 1.000 → neutral
- **R6a Restoration:** 1.041 ↑ amplifies
- **R6b Seismic:** 1.020 ↑ amplifies
- **R7 Digital Readiness:** 1.030 ↑ amplifies

This month's spotlight — automatically selected for September 2026 — is **Fürstenberg** in DE40A, Oberhavel. With an R_final of 0.646 (High band, 64th fleet percentile), its risk profile is dominated by the T (Transition) component at 789% of its weight ceiling, followed by E (Economic) at 337%. Modifiers collectively shift R_base by 103.4%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

## Cyber & Economic Exposure Monitor

**Why this section exists:** Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

*R7 v2 CRA-anchored transition (Q3 2026 → Q1 2027 dual-write):* the R7 signal is entering a dual-write window where the legacy R7_cyber (DESI + ACN scalar proxy) co-exists with R7_cyber_v2 (CRA Article 14 + NIS2 Article 21 register-anchored composite, Path C+D). Consumers may select either; substation audit trail carries `_r7_cyber_v2_source` when v2 populated.

CYBER HIGH EXPOSURE

**0**

0.0% of fleet

BLIND SPOTS

**21866**

High/Critical/Extreme + R7 < median

AVG R7 MODIFIER

**1.0200**

Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

**0**

0.0% · R3 ≥ 1.14

## B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.


• Low • Medium • High • Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **21866 blind-spot substations** — scoring High, Critical, or Extreme on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0199). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Unknown (395), Paderborn (372), Berlin (364)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

## B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

### 🏭 Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Substations serving highest-value economic activity — pharma / chemicals / data centres / financial districts / heavy industry

**10,806** substations (10.0%) High/Crit/Ext: **5737** (53.1%) Avg R: **0.640** Avg E: **0.445** / 0.10

Low 2135 Med 2934 High 3103 Crit 1940 Ext 694

### 🔧 Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh

Industrial belt — significant continuity requirements

**16,206** substations (15.0%) High/Crit/Ext: **8304** (51.2%) Avg R: **0.636** Avg E: **0.418** / 0.10

Low 3355 Med 4547 High 4554 Crit 2767 Ext 983

### 🏪 Commercial / SME-Dense

Est. VoLL: €3–8/kWh

Service economy, retail, urban commercial districts — moderate individual impact

**27,008** substations (25.0%) High/Crit/Ext: **13359** (49.5%) Avg R: **0.631** Avg E: **0.376** / 0.10

Low 5975 Med 7674 High 7509 Crit 4350 Ext 1500

### 🌾 Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse activity, agricultural land — lower VoLL

**53,996** substations (50.0%) High/Crit/Ext: **25024** (46.3%) Avg R: **0.622** Avg E: **0.298** / 0.10

Low 13350 Med 15622 High 15249 Crit 7547 Ext 2228

**Value of Lost Load (VoLL) context:** ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's Economic component (CVIESTR·E) segments substations by the economic value they serve. **5737 substations** combine capital-intensive economic fabric (top-decile Economic component (E ≥ 0.434, P90+)) with High, Critical, or Extreme risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Unknown (192), Paderborn (189), Berlin (175)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 10.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

## B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify provinces combining digital exposure with high grid stress.


WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

| | | |
|---|---|---|
| 1 | DE713 ⚠ high R | 1.0155 |
| 2 | DEG0S | 1.0156 |
| 3 | DE403 ⚠ high R | 1.0164 |
| 4 | DEG0D | 1.0172 |
| 5 | DE272 | 1.0173 |
| 6 | DE277 | 1.0175 |
| 7 | DEB3A | 1.0178 |
| 8 | DEG0J | 1.0182 |
| 9 | DEB11 | 1.0182 |
| 10 | DEA12 ⚠ high R | 1.0182 |
| 11 | DEF01 ⚠ high R | 1.0183 |
| 12 | DEA24 | 1.0184 |
| 13 | DE274 | 1.0184 |
| 14 | DEG0V | 1.0185 |
| 15 | DE912 ⚠ high R | 1.0185 |

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **DE713** has the lowest average R7 (1.0155), while **DE233** leads at 1.0222 — a spread of 0.0067 across the R7 range. **94 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

## B.4 Monthly Pulse

**Inaugural edition — Baseline Snapshot (September 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0200, median = 1.0199; 0 substations classified HIGH cyber-exposure; 21866 blind-spot substations (High/Critical/Extreme risk + below-median R7); 7458 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

# Data Refresh & Changelog

The SSI v4.0.2 draws on **— verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

—

— variables

HOURLY — MONTHLY

0

High-frequency feeds

QUARTERLY / ANNUAL

0

Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

## Data Source Registry — September 2026

| | | | | |
|---|---|---|---|---|
| OSM | OSM Power Infrastructure | WEEKLY | Node/edge | 3 vars |
| DWD | DWD (Deutscher Wetterdienst) | DAILY | ~1 km | 3 vars |
| BBK | BBK Bevölkerungsschutz (Civil Protection) | CONTINUOUS | Landkreis | 1 var |
| DESTKFZ | DESTATIS KFZ-Bestand | QUARTERLY | Landkreis | 1 var |
| BMUV | BMUV Umwelt + Verbraucher | QUARTERLY | Landkreis | 1 var |
| BNETZ | BNetzA (Bundesnetzagentur) | ANNUAL | Landkreis | 8 vars |
| 50HZ | 50Hertz Netzentwicklungsplan | ANNUAL | Control area NE | 2 vars |
| AMPR | Amprion Netzentwicklungsplan | ANNUAL | Control area W | 2 vars |
| TENN | TenneT TSO DE Netzentwicklungsplan | ANNUAL | Control area N-S | 2 vars |
| TRBW | TransnetBW Netzentwicklungsplan | ANNUAL | Control area SW | 1 var |
| DESTAT | DESTATIS (Statistisches Bundesamt) | ANNUAL | Landkreis (NUTS-3) | 8 vars |
| UBA | UBA Umweltbundesamt | ANNUAL | Landkreis | 4 vars |
| BFG | BfG Bundesanstalt für Gewässerkunde | ANNUAL | Landkreis | 2 vars |
| LFUBY | LfU Bayern (Landesamt für Umwelt) | ANNUAL | Bavarian LK | 1 var |
| JRC | JRC DSO Observatory | ANNUAL | DSO-level | 2 vars |
| EEA | EEA Air Quality e-Reporting | ANNUAL | ~1 km + station | 3 vars |
| EURO | Eurostat Energy Statistics | ANNUAL | NUTS2 | 3 vars |
| DESI | DESI Digital Economy Index | ANNUAL | EU Regional | 2 vars |
| BMWK | BMWK Wirtschaft + Klimaschutz | ANNUAL | Bundesland | 2 vars |
| BMU | BMU Umwelt + Naturschutz | ANNUAL | Bundesland | 1 var |
| BMF | BMF Bundesfinanzministerium | ANNUAL | Landkreis | 1 var |
| BSI | BSI Sicherheit Informationstechnik | ANNUAL | KRITIS asset | 2 vars |
| BBANK | Bundesbank (Deutsche Bundesbank) | ANNUAL | Regional | 2 vars |
| DENA | dena (Deutsche Energie-Agentur) | ANNUAL | Bundesland | 2 vars |

| | | | | |
|---|---|---|---|---|
| BBSR | BBSR Raumordnung | ANNUAL | Landkreis | 2 vars |
| BNMON | BNetzA Monitoringbericht | ANNUAL | DSO-level | 2 vars |
| GOVDE | GovData.de (federal open data) | ANNUAL | Landkreis | 1 var |
| DIW | DIW Berlin (Wirtschaftsforschung) | ANNUAL | Bundesland | 1 var |
| BMI | BMI Bundesinnenministerium (KRITIS coord) | ANNUAL | KRITIS asset | 1 var |
| BGR | BGR Geowissenschaften und Rohstoffe | STATIC | Landkreis | 3 vars |
| COPER | Copernicus CDS / ERA5 | STATIC | 0.25° (~25 km) | 4 vars |
| IEEE-1 | IEEE C57.91 / IEC 60076 | STATIC | Asset-level | 16 vars |
| AGS | AGS Amtlicher Gemeindeschlüssel | STATIC | Gemeinde (8-digit) | 1 var |
| ISO-9223 | ISO 9223 Corrosion | DERIVED | Landkreis | 1 var |
| ENTSE | ENTSO-E Transparency | HOURLY | Bidding zone | 2 vars |

## C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

## C.2 Fleet Score Snapshot

FLEET SIZE
**108,016**
substations scored

MEDIAN R
**0.627**
P5=0.477 · P95=0.784

HIGH + CRITICAL + EXTREME
**52424**
48.5% of fleet

HIGH CONFIDENCE
**99%**
106502 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 24815 (23.0%) Medium 30777 (28.5%) High 30415 (28.2%) Critical 16604 (15.4%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **108,016 substation scores remain unchanged** this edition. The fleet median R of 0.627 (mean 0.628) reflects a distribution skewed toward the lower bands, with 23.0% in Low and 28.5% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

## C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources ( IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across P0, P1, and P2 releases

| | | | |
|---|---|---|---|
| V42-1 | NEW | R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention) | §5 |
| V42-2 | NEW | R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20]) | §5 |
| V42-3 | NEW | R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02]) | §5 |
| V42-4 | NEW | R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05]) | §5 |
| V42-5 | NEW | R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10}) | §5 |
| V42-6 | NEW | R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12]) | §5 |
| V42-7 | NEW | Re composite — bounded resilience scalar Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c - 1.00); Re_norm = clip((Re_raw - 0.920)/(1.787 - 0.920), 0, 1) | §7 |
| V42-8 | NEW | R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring) | §14 |
| V42-9 | NEW | W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6) | §4 |
| V42-10 | CHANGE | 95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers) | data |
| V42-11 | CHANGE | Substation fleet: 13,251 substations (per OSM canonical extract + national TSO/DSO registry cross-validation) | data |

SECTION D — MONTHLY DEEP-DIVE

# The Bavaria Solar Corridor

> **Deep-dive rotation:** Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

## D.1 Theme Context

THEME-RELEVANT SUBSTATIONS

**2**

0 EHV · 2 HV

HIGH BAND

**0 + 1**

50.0% of corridor

CORRIDOR MEDIAN R

**0.799**

Fleet median: 0.627

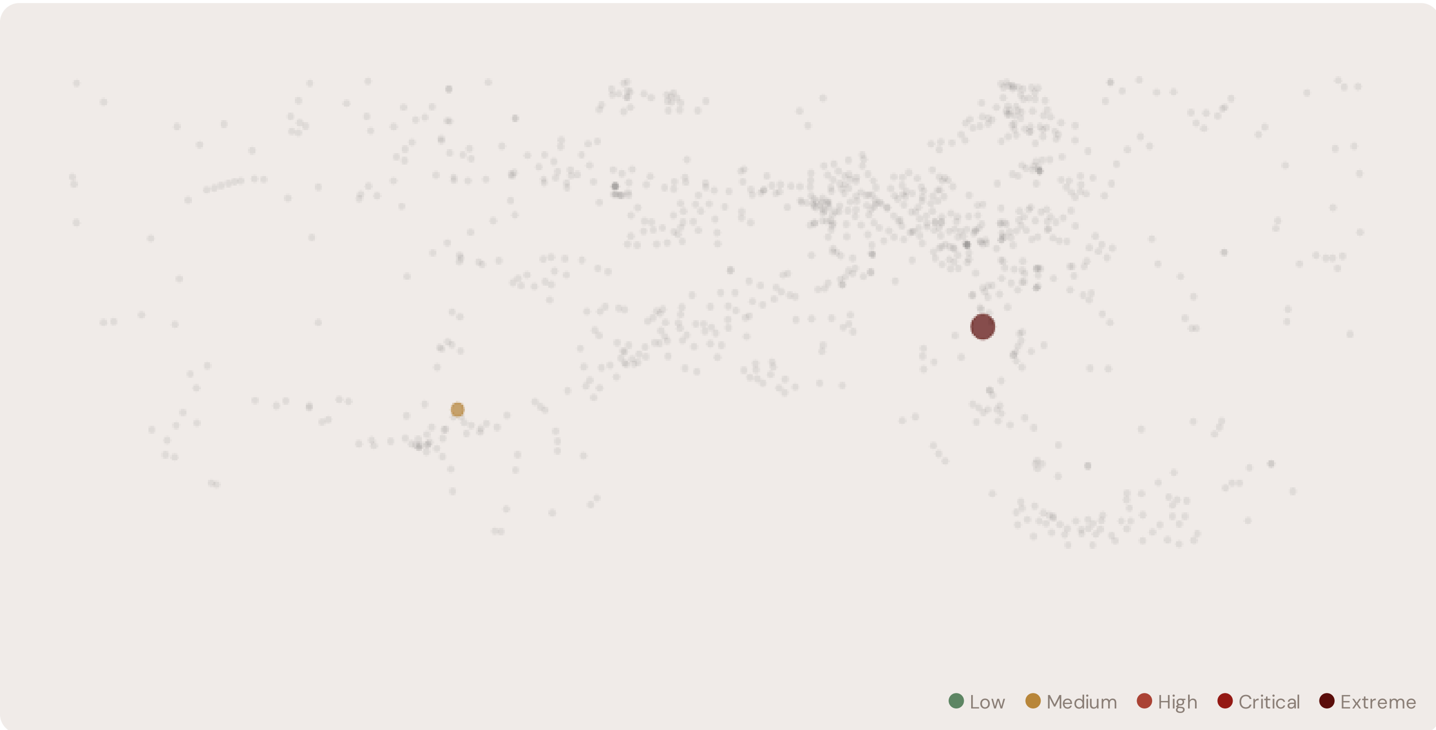
WORST PROVINCE

**unknown**

Avg R: 0.695 · 2 substations

The Saarland hosts **2 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **0 substations (0.0%)** are classified High, with only **0** in the Low band. 50% of corridor substations score above the national fleet median of 0.627. The corridor median R of 0.799 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

## D.2 Corridor Map and Scoring




| SUBSTATION | PROVINCE | R_FINAL | BAND | C | V | I | E | S | T | RE V4.2 | ALERT |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Substation 11000059027 | | 0.799 | Extreme | 0.383 | 0.363 | 0.411 | 0.318 | 0.443 | 0.305 | | |
| Substation 11000038382 | | 0.592 | Medium | 0.398 | 0.413 | 0.318 | 0.263 | 0.453 | 0.428 | | |

*… -8 additional substations — explore full corridor in Digital Twin →*

## D.3 Component Analysis

| COMPONENT AVERAGES — SAARLAND VS FLEET | |
|---|---|
| C — Continuity | 0.391 vs 0.350 (+12%) |
| I — Infrastructure | 0.365 vs 0.350 (+4%) |
| S — Saturation | 0.448 vs 0.350 (+28%) |
| V — Voltage | 0.388 vs 0.350 (+11%) |
| E — Economic | 0.290 vs 0.350 (-17%) |
| T — Transition | 0.367 vs 0.350 (+5%) |

| MODIFIER AVERAGES — SAARLAND VS FLEET | | |
|---|---|---|
| R3 Consequence | 0.972 | fleet 0.972 ↓ |
| R4 Graph Crit. | 0.994 | fleet 0.980 → |
| R6a Restoration | 1.034 | fleet 1.020 ↑ |
| R6b Seismic | 1.014 | fleet 1.000 ↑ |
| R7 Digital Read. | 1.027 | fleet 1.020 ↑ |

The dominant risk driver across the Saarland corridor is **T (Transition)**, averaging 0.367 against a fleet average of 0.350 — occupying 733% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.388 vs fleet 0.350 (388% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.972 (fleet: 0.972), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.014, capturing the local seismic exposure — a dimension invisible to every competing framework.

## D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

**unknown** 2 subs · avg R = 0.695 · H:0 M:0

The region regions-level breakdown reveals significant intra-regional variation. **unknown** leads with a mean R of 0.695 across 2 substations, while **unknown** scores 0.695 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Saarland is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

## D.5 Implications

**1 Saarland substations** score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the unknown sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

# OECD Context Card

**How this works:** Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on Germany's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

OECD CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · August 2026

### Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.

| | |
| --- | --- |
| 🇩🇪 Germany | 11.7 |
| 🇨🇭 Switzerland | ~15 |
| 🇳🇱 Netherlands | 23.9 |
| 🇫🇷 France | ~36 |
| 🇬🇧 UK | ~40 |
| 🇮🇹 Italy | ~43 |
| 🇪🇸 Spain | ~48 |
| 🇭🇺 Hungary | ~57 |
| 🇵🇱 Poland | ~95 |
| 🇷🇴 Romania | ~290 |

Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

**This month's key insight:** The Saarland corridor deep-dive shows **2 substations** (of 2) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.391 is **1.1× the fleet average** of 0.350. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

## Bundesländer Risk Ranking — All 409 German Bundesländer

Where does this corridor sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

| | | |
| --- | --- | --- |
| #1 | **Saarland** | 2 subs · avg R = 0.695 · H:0 |
| #2 | Schleswig-Holstein | 8 subs · avg R = 0.695 · H:2 C:4 |
| #3 | Sachsen | 3 subs · avg R = 0.679 · H:1 C:1 |
| #4 | Rheinland-Pfalz | 3 subs · avg R = 0.676 · H:0 C:1 |
| #5 | Offenbach am Main, Kreisfreie Stadt | 367 subs · avg R = 0.669 · H:279 C:13 |
| #6 | Pirmasens, Kreisfreie Stadt | 12 subs · avg R = 0.667 · H:3 C:6 |
| #7 | Aurich | 38 subs · avg R = 0.656 · H:8 C:10 |
| #8 | Emden, Kreisfreie Stadt | 22 subs · avg R = 0.656 · H:7 C:2 |
| #9 | Chemnitz, Kreisfreie Stadt | 112 subs · avg R = 0.654 · H:28 C:36 |

| | | |
|---|---|---|
| #10 | Neu-Ulm | 148 subs · avg R = 0.652 · H:40 C:33 |
| #11 | Wolfsburg, Kreisfreie Stadt | 60 subs · avg R = 0.652 · H:18 C:12 |
| #12 | Baden-Württemberg | 2 subs · avg R = 0.651 · H:2 |
| #13 | Bremerhaven, Kreisfreie Stadt | 12 subs · avg R = 0.649 · H:4 C:1 |
| #14 | Passau, Kreisfreie Stadt | 19 subs · avg R = 0.648 · H:4 C:4 |
| #15 | Wittenberg | 109 subs · avg R = 0.647 · H:34 C:21 |
| #16 | Donnersbergkreis | 79 subs · avg R = 0.647 · H:25 C:15 |
| #17 | Hamm, Kreisfreie Stadt | 53 subs · avg R = 0.646 · H:15 C:9 |
| #18 | Schwarzwald-Baar-Kreis | 177 subs · avg R = 0.646 · H:54 C:22 |
| #19 | Merzig-Wadern | 130 subs · avg R = 0.646 · H:36 C:21 |
| #20 | Grafschaft Bentheim | 66 subs · avg R = 0.645 · H:17 C:13 |
| #21 | Vulkaneifel | 71 subs · avg R = 0.645 · H:24 C:7 |
| #22 | Offenbach, Landkreis | 797 subs · avg R = 0.645 · H:322 C:126 |
| #23 | Aschaffenburg, Kreisfreie Stadt | 75 subs · avg R = 0.644 · H:25 C:12 |
| #24 | Lindau (Bodensee) | 63 subs · avg R = 0.643 · H:24 C:13 |
| #25 | Straubing, Kreisfreie Stadt | 23 subs · avg R = 0.643 · H:8 C:2 |
| #26 | Landshut, Kreisfreie Stadt | 40 subs · avg R = 0.642 · H:12 C:9 |
| #27 | Nordwestmecklenburg | 212 subs · avg R = 0.642 · H:59 C:49 |
| #28 | Hameln-Pyrmont | 63 subs · avg R = 0.642 · H:15 C:11 |
| #29 | Schwerin, Kreisfreie Stadt | 64 subs · avg R = 0.642 · H:17 C:12 |
| #30 | Aichach-Friedberg | 228 subs · avg R = 0.641 · H:58 C:55 |
| #31 | Rheinisch-Bergischer Kreis | 124 subs · avg R = 0.640 · H:37 C:19 |
| #32 | Hohenlohekreis | 152 subs · avg R = 0.640 · H:50 C:28 |
| #33 | Kusel | 82 subs · avg R = 0.640 · H:24 C:16 |
| #34 | Mülheim an der Ruhr, Kreisfreie Stadt | 71 subs · avg R = 0.640 · H:24 C:6 |
| #35 | Nordsachsen | 210 subs · avg R = 0.640 · H:72 C:33 |
| #36 | Saalfeld-Rudolstadt | 209 subs · avg R = 0.639 · H:61 C:38 |
| #37 | Uelzen | 186 subs · avg R = 0.639 · H:49 C:30 |
| #38 | Bottrop, Kreisfreie Stadt | 39 subs · avg R = 0.639 · H:15 C:5 |
| #39 | Stuttgart, Stadtkreis | 212 subs · avg R = 0.639 · H:78 C:28 |

#40 Celle — 91 subs · avg R = 0.639 · H:28 C:21

#41 Hochsauerlandkreis — 104 subs · avg R = 0.638 · H:29 C:13

#42 Hersfeld-Rotenburg — 338 subs · avg R = 0.638 · H:91 C:65

#43 Salzlandkreis — 209 subs · avg R = 0.638 · H:85 C:22

#44 Wilhelmshaven, Kreisfreie Stadt — 80 subs · avg R = 0.638 · H:22 C:13

#45 Wesel — 141 subs · avg R = 0.638 · H:40 C:26

#46 Heilbronn, Landkreis — 375 subs · avg R = 0.638 · H:122 C:73

#47 Frankfurt am Main, Kreisfreie Stadt — 242 subs · avg R = 0.638 · H:81 C:41

#48 Starnberg — 146 subs · avg R = 0.637 · H:39 C:29

#49 Ostprignitz-Ruppin — 74 subs · avg R = 0.637 · H:22 C:12

#50 Barnim — 108 subs · avg R = 0.637 · H:29 C:16

#51 Augsburg, Landkreis — 286 subs · avg R = 0.637 · H:86 C:53

#52 Vogtlandkreis — 94 subs · avg R = 0.637 · H:33 C:17

#53 Warendorf — 191 subs · avg R = 0.637 · H:55 C:36

#54 Rosenheim, Landkreis — 457 subs · avg R = 0.637 · H:121 C:84

#55 Schwabach, Kreisfreie Stadt — 26 subs · avg R = 0.637 · H:10 C:5

#56 Mecklenburgische Seenplatte — 514 subs · avg R = 0.636 · H:167 C:85

#57 Mecklenburg-Vorpommern — 5 subs · avg R = 0.636 · H:1 C:2

#58 Jena, Kreisfreie Stadt — 87 subs · avg R = 0.636 · H:23 C:13

#59 Oder-Spree — 208 subs · avg R = 0.636 · H:97 C:16

#60 Kaiserslautern, Landkreis — 121 subs · avg R = 0.636 · H:38 C:18

#61 Cochem-Zell — 147 subs · avg R = 0.636 · H:54 C:18

#62 Oberallgäu — 397 subs · avg R = 0.636 · H:97 C:78

#63 Bonn, Kreisfreie Stadt — 343 subs · avg R = 0.636 · H:91 C:66

#64 Freudenstadt — 104 subs · avg R = 0.636 · H:37 C:18

#65 Altenkirchen (Westerwald) — 136 subs · avg R = 0.636 · H:39 C:24

#66 Oberspreewald-Lausitz — 200 subs · avg R = 0.636 · H:67 C:38

#67 Konstanz — 278 subs · avg R = 0.636 · H:90 C:49

#68 Wuppertal, Kreisfreie Stadt — 267 subs · avg R = 0.636 · H:67 C:49

#69 Solingen, Kreisfreie Stadt — 45 subs · avg R = 0.636 · H:12 C:8

#70 Altmarkkreis Salzwedel — 281 subs · avg R = 0.635 · H:74 C:42

#71 Alzey-Worms — 106 subs · avg R = 0.635 · H:33 C:20

#72 Zwickau — 141 subs · avg R = 0.635 · H:32 C:27

#73 Ulm, Stadtkreis — 58 subs · avg R = 0.635 · H:17 C:11

#74 Nürnberger Land — 296 subs · avg R = 0.635 · H:79 C:58

#75 Meißen — 448 subs · avg R = 0.635 · H:128 C:83

#76 Rosenheim, Kreisfreie Stadt — 129 subs · avg R = 0.635 · H:51 C:14

#77 Städteregion Aachen — 553 subs · avg R = 0.635 · H:143 C:103

#78 Stormarn — 413 subs · avg R = 0.635 · H:112 C:74

#79 Erfurt, Kreisfreie Stadt — 335 subs · avg R = 0.635 · H:84 C:69

#80 Osterholz — 538 subs · avg R = 0.635 · H:149 C:88

#81 Siegen-Wittgenstein — 161 subs · avg R = 0.635 · H:46 C:38

#82 Schwalm-Eder-Kreis — 413 subs · avg R = 0.635 · H:105 C:73

#83 Soest — 289 subs · avg R = 0.635 · H:81 C:51

#84 Höxter — 242 subs · avg R = 0.634 · H:72 C:38

#85 Neustadt a. d. Aisch-Bad Windsheim — 314 subs · avg R = 0.634 · H:96 C:47

#86 Bad Kissingen — 238 subs · avg R = 0.634 · H:64 C:44

#87 Oldenburg (Oldenburg), Kreisfreie Stadt — 78 subs · avg R = 0.634 · H:20 C:14

#88 Neustadt a. d. Waldnaab — 426 subs · avg R = 0.634 · H:125 C:69

#89 Weißenburg-Gunzenhausen — 220 subs · avg R = 0.634 · H:60 C:39

#90 Miesbach — 228 subs · avg R = 0.634 · H:58 C:40

#91 Rendsburg-Eckernförde — 245 subs · avg R = 0.633 · H:74 C:43

#92 Gifhorn — 377 subs · avg R = 0.633 · H:109 C:49

#93 Weiden i. d. Opf, Kreisfreie Stadt — 27 subs · avg R = 0.633 · H:11 C:3

#94 Mayen-Koblenz — 145 subs · avg R = 0.633 · H:42 C:17

#95 Eifelkreis Bitburg-Prüm — 84 subs · avg R = 0.633 · H:23 C:14

#96 Vorpommern-Rügen — 396 subs · avg R = 0.633 · H:107 C:65

#97 Bodenseekreis — 319 subs · avg R = 0.633 · H:85 C:55

#98 Ludwigsburg — 336 subs · avg R = 0.633 · H:95 C:50

#99 Märkisch-Oderland — 142 subs · avg R = 0.633 · H:39 C:25

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|---|---|
| #100 Lörrach | 165 subs · avg R = 0.633 · H:41 C:22 |
| #101 Diepholz | 272 subs · avg R = 0.633 · H:71 C:49 |
| #102 Miltenberg | 252 subs · avg R = 0.633 · H:69 C:49 |
| #103 Nordhausen | 143 subs · avg R = 0.633 · H:39 C:34 |
| #104 Neckar-Odenwald-Kreis | 245 subs · avg R = 0.633 · H:66 C:47 |
| #105 Weimarer Land | 69 subs · avg R = 0.632 · H:20 C:8 |
| #106 Düren | 150 subs · avg R = 0.632 · H:40 C:22 |
| #107 Pinneberg | 307 subs · avg R = 0.632 · H:89 C:47 |
| #108 Mainz-Bingen | 117 subs · avg R = 0.632 · H:35 C:19 |
| #109 Bremen, Kreisfreie Stadt | 288 subs · avg R = 0.632 · H:86 C:37 |
| #110 Rhein-Hunsrück-Kreis | 67 subs · avg R = 0.632 · H:18 C:12 |
| #111 Freyung-Grafenau | 252 subs · avg R = 0.632 · H:68 C:45 |
| #112 Augsburg, Kreisfreie Stadt | 283 subs · avg R = 0.632 · H:76 C:52 |
| #113 Bamberg, Kreisfreie Stadt | 129 subs · avg R = 0.632 · H:32 C:22 |
| #114 Flensburg, Kreisfreie Stadt | 16 subs · avg R = 0.632 · H:4 C:3 |
| #115 Ludwigslust-Parchim | 286 subs · avg R = 0.632 · H:90 C:44 |
| #116 Regionalverband Saarbrücken | 684 subs · avg R = 0.632 · H:206 C:109 |
| #117 Main-Taunus-Kreis | 250 subs · avg R = 0.632 · H:76 C:39 |
| #118 Esslingen | 396 subs · avg R = 0.632 · H:105 C:56 |
| #119 Mönchengladbach, Kreisfreie Stadt | 483 subs · avg R = 0.632 · H:135 C:84 |
| #120 Vorpommern-Greifswald | 576 subs · avg R = 0.632 · H:156 C:102 |
| #121 Oberhavel | 163 subs · avg R = 0.632 · H:44 C:25 |
| #122 Ansbach, Kreisfreie Stadt | 137 subs · avg R = 0.631 · H:37 C:27 |
| #123 Rotenburg (Wümme) | 141 subs · avg R = 0.631 · H:56 C:16 |
| #124 Calw | 157 subs · avg R = 0.631 · H:45 C:23 |
| #125 Rottal-Inn | 512 subs · avg R = 0.631 · H:160 C:70 |
| #126 Schwandorf | 421 subs · avg R = 0.631 · H:137 C:54 |
| #127 Magdeburg, Kreisfreie Stadt | 189 subs · avg R = 0.631 · H:49 C:31 |
| #128 Neumünster, Kreisfreie Stadt | 24 subs · avg R = 0.631 · H:4 C:4 |
| #129 Elbe-Elster | 87 subs · avg R = 0.631 · H:20 C:16 |

#130 Rastatt — 475 subs · avg R = 0.631 · H:143 C:84

#131 Rhein-Neckar-Kreis — 802 subs · avg R = 0.631 · H:228 C:129

#132 Ostholstein — 470 subs · avg R = 0.631 · H:143 C:61

#133 Werra-Meißner-Kreis — 220 subs · avg R = 0.631 · H:69 C:34

#134 Rhein-Kreis Neuss — 1020 subs · avg R = 0.631 · H:288 C:163

#135 Nürnberg, Kreisfreie Stadt — 602 subs · avg R = 0.631 · H:179 C:95

#136 Minden-Lübbecke — 196 subs · avg R = 0.631 · H:47 C:33

#137 Straubing-Bogen — 384 subs · avg R = 0.631 · H:125 C:57

#138 Salzgitter, Kreisfreie Stadt — 33 subs · avg R = 0.631 · H:7 C:5

#139 Cham — 383 subs · avg R = 0.631 · H:106 C:54

#140 Frankfurt (Oder), Kreisfreie Stadt — 24 subs · avg R = 0.631 · H:9

#141 Unknown — 1933 subs · avg R = 0.631 · H:565 C:304

#142 Reutlingen — 299 subs · avg R = 0.631 · H:88 C:40

#143 Main-Kinzig-Kreis — 1225 subs · avg R = 0.631 · H:333 C:183

#144 Saarlouis — 231 subs · avg R = 0.630 · H:87 C:26

#145 Holzminden — 62 subs · avg R = 0.630 · H:17 C:11

#146 Lüneburg, Landkreis — 210 subs · avg R = 0.630 · H:54 C:40

#147 Rhein-Erft-Kreis — 166 subs · avg R = 0.630 · H:46 C:25

#148 Germersheim — 71 subs · avg R = 0.630 · H:17 C:12

#149 Südliche Weinstraße — 172 subs · avg R = 0.630 · H:64 C:21

#150 Oberbergischer Kreis — 162 subs · avg R = 0.630 · H:44 C:21

#151 Teltow-Fläming — 361 subs · avg R = 0.630 · H:102 C:65

#152 Bautzen — 373 subs · avg R = 0.630 · H:113 C:63

#153 Weilheim-Schongau — 114 subs · avg R = 0.630 · H:28 C:21

#154 Göppingen — 201 subs · avg R = 0.630 · H:60 C:33

#155 Frankenthal (Pfalz), Kreisfreie Stadt — 83 subs · avg R = 0.630 · H:15 C:14

#156 Delmenhorst, Kreisfreie Stadt — 103 subs · avg R = 0.630 · H:26 C:13

#157 Limburg-Weilburg — 459 subs · avg R = 0.630 · H:127 C:72

#158 Goslar — 168 subs · avg R = 0.630 · H:47 C:23

#159 St. Wendel — 118 subs · avg R = 0.630 · H:29 C:26

#160 Südwestpfalz — 161 subs · avg R = 0.630 · H:49 C:23

#161 Ennepe-Ruhr-Kreis — 717 subs · avg R = 0.630 · H:185 C:106

#162 Birkenfeld — 65 subs · avg R = 0.630 · H:21 C:12

#163 Remscheid, Kreisfreie Stadt — 65 subs · avg R = 0.630 · H:21 C:6

#164 Eichstätt — 421 subs · avg R = 0.630 · H:112 C:66

#165 Bochum, Kreisfreie Stadt — 678 subs · avg R = 0.629 · H:194 C:121

#166 Hildesheim — 216 subs · avg R = 0.629 · H:61 C:37

#167 Mühldorf a. Inn — 85 subs · avg R = 0.629 · H:19 C:14

#168 Dortmund, Kreisfreie Stadt — 130 subs · avg R = 0.629 · H:39 C:17

#169 Osnabrück, Kreisfreie Stadt — 12 subs · avg R = 0.629 · H:3 C:2

#170 Günzburg — 116 subs · avg R = 0.629 · H:28 C:19

#171 Ammerland — 67 subs · avg R = 0.629 · H:18 C:11

#172 Cottbus, Kreisfreie Stadt — 117 subs · avg R = 0.629 · H:34 C:15

#173 Lübeck, Kreisfreie Stadt — 552 subs · avg R = 0.629 · H:149 C:89

#174 Erlangen, Kreisfreie Stadt — 188 subs · avg R = 0.629 · H:54 C:28

#175 Bielefeld, Kreisfreie Stadt — 350 subs · avg R = 0.629 · H:102 C:57

#176 Altenburger Land — 120 subs · avg R = 0.629 · H:30 C:22

#177 Bayreuth, Landkreis — 197 subs · avg R = 0.629 · H:51 C:36

#178 Westerwaldkreis — 495 subs · avg R = 0.629 · H:122 C:89

#179 Uckermark — 217 subs · avg R = 0.629 · H:55 C:39

#180 Donau-Ries — 149 subs · avg R = 0.629 · H:39 C:23

#181 Wolfenbüttel — 141 subs · avg R = 0.629 · H:41 C:20

#182 Fürth, Kreisfreie Stadt — 92 subs · avg R = 0.629 · H:26 C:8

#183 Leipzig, Kreisfreie Stadt — 1004 subs · avg R = 0.629 · H:272 C:182

#184 Unstrut-Hainich-Kreis — 80 subs · avg R = 0.629 · H:17 C:19

#185 Berchtesgadener Land — 323 subs · avg R = 0.629 · H:92 C:52

#186 Bad Kreuznach — 86 subs · avg R = 0.629 · H:21 C:11

#187 Peine — 54 subs · avg R = 0.629 · H:14 C:9

#188 Bamberg, Landkreis — 329 subs · avg R = 0.629 · H:81 C:53

#189 Karlsruhe, Landkreis — 947 subs · avg R = 0.629 · H:265 C:148

#190 Nordfriesland 305 subs · avg R = 0.629 · H:91 C:45

#191 Main-Spessart 184 subs · avg R = 0.629 · H:55 C:28

#192 Schmalkalden-Meiningen 146 subs · avg R = 0.629 · H:41 C:24

#193 Mannheim, Stadtkreis 644 subs · avg R = 0.629 · H:170 C:103

#194 Landshut, Landkreis 520 subs · avg R = 0.629 · H:158 C:66

#195 Groß-Gerau 505 subs · avg R = 0.629 · H:146 C:72

#196 Ahrweiler 70 subs · avg R = 0.629 · H:19 C:7

#197 Regen 185 subs · avg R = 0.628 · H:45 C:33

#198 Amberg-Sulzbach 479 subs · avg R = 0.628 · H:133 C:66

#199 Duisburg, Kreisfreie Stadt 120 subs · avg R = 0.628 · H:36 C:18

#200 Dresden, Kreisfreie Stadt 688 subs · avg R = 0.628 · H:193 C:109

#201 Gera, Kreisfreie Stadt 41 subs · avg R = 0.628 · H:14 C:5

#202 Gütersloh 1052 subs · avg R = 0.628 · H:278 C:158

#203 Alb-Donau-Kreis 418 subs · avg R = 0.628 · H:113 C:58

#204 Erzgebirgskreis 216 subs · avg R = 0.628 · H:60 C:33

#205 Vogelsbergkreis 639 subs · avg R = 0.628 · H:179 C:97

#206 Trier, Kreisfreie Stadt 122 subs · avg R = 0.628 · H:33 C:21

#207 Schweinfurt, Kreisfreie Stadt 179 subs · avg R = 0.628 · H:47 C:34

#208 Dahme-Spreewald 319 subs · avg R = 0.628 · H:81 C:62

#209 Mettmann 346 subs · avg R = 0.628 · H:103 C:58

#210 Kempten (Allgäu), Kreisfreie Stadt 151 subs · avg R = 0.628 · H:45 C:23

#211 Oldenburg, Landkreis 312 subs · avg R = 0.628 · H:90 C:45

#212 Göttingen 1175 subs · avg R = 0.628 · H:331 C:173

#213 Wunsiedel i. Fichtelgebirge 126 subs · avg R = 0.628 · H:42 C:12

#214 Neumarkt i. d. OPf. 506 subs · avg R = 0.628 · H:143 C:80

#215 Coesfeld 304 subs · avg R = 0.628 · H:86 C:48

#216 Spree-Neiße 115 subs · avg R = 0.628 · H:27 C:19

#217 Emmendingen 186 subs · avg R = 0.628 · H:55 C:28

#218 Deggendorf 514 subs · avg R = 0.628 · H:137 C:76

#219 Tübingen, Landkreis 415 subs · avg R = 0.628 · H:118 C:55

#220 Fulda 671 subs · avg R = 0.628 · H:183 C:95

#221 Schweinfurt, Landkreis 427 subs · avg R = 0.628 · H:140 C:60

#222 München, Landkreis 1011 subs · avg R = 0.628 · H:267 C:181

#223 Viersen 998 subs · avg R = 0.628 · H:272 C:159

#224 Wartburgkreis 150 subs · avg R = 0.628 · H:37 C:16

#225 Landsberg am Lech 149 subs · avg R = 0.627 · H:43 C:23

#226 Dachau 332 subs · avg R = 0.627 · H:87 C:52

#227 Bernkastel-Wittlich 218 subs · avg R = 0.627 · H:73 C:33

#228 Hagen, Kreisfreie Stadt 174 subs · avg R = 0.627 · H:39 C:30

#229 Münster, Kreisfreie Stadt 759 subs · avg R = 0.627 · H:225 C:118

#230 Berlin 1802 subs · avg R = 0.627 · H:487 C:292

#231 Regensburg, Landkreis 504 subs · avg R = 0.627 · H:139 C:71

#232 Fürstenfeldbruck 184 subs · avg R = 0.627 · H:56 C:23

#233 Unterallgäu 59 subs · avg R = 0.627 · H:18 C:9

#234 Mittelsachsen 556 subs · avg R = 0.627 · H:168 C:81

#235 Darmstadt-Dieburg 557 subs · avg R = 0.627 · H:161 C:72

#236 Dithmarschen 662 subs · avg R = 0.627 · H:171 C:94

#237 Rems-Murr-Kreis 426 subs · avg R = 0.627 · H:115 C:72

#238 Ortenaukreis 678 subs · avg R = 0.627 · H:186 C:103

#239 Herne, Kreisfreie Stadt 63 subs · avg R = 0.627 · H:10 C:9

#240 Altötting 178 subs · avg R = 0.627 · H:48 C:23

#241 Hamburg 405 subs · avg R = 0.627 · H:110 C:68

#242 Potsdam, Kreisfreie Stadt 560 subs · avg R = 0.627 · H:150 C:91

#243 Dingolfing-Landau 318 subs · avg R = 0.627 · H:81 C:57

#244 Havelland 213 subs · avg R = 0.627 · H:54 C:35

#245 Paderborn 1982 subs · avg R = 0.627 · H:571 C:282

#246 Köln, Kreisfreie Stadt 980 subs · avg R = 0.627 · H:273 C:136

#247 Heinsberg 776 subs · avg R = 0.627 · H:217 C:117

#248 Böblingen 377 subs · avg R = 0.627 · H:115 C:59

#249 Freiburg im Breisgau, Stadtkreis 357 subs · avg R = 0.627 · H:85 C:67

#250 Saalekreis — 216 subs · avg R = 0.627 · H:69 C:29

#251 Ravensburg — 338 subs · avg R = 0.627 · H:108 C:46

#252 Rheingau-Taunus-Kreis — 561 subs · avg R = 0.626 · H:153 C:88

#253 Garmisch-Partenkirchen — 121 subs · avg R = 0.626 · H:31 C:14

#254 Freising — 389 subs · avg R = 0.626 · H:110 C:56

#255 Baden-Baden, Stadtkreis — 181 subs · avg R = 0.626 · H:43 C:29

#256 Kleve — 198 subs · avg R = 0.626 · H:59 C:24

#257 Speyer, Kreisfreie Stadt — 64 subs · avg R = 0.626 · H:17 C:9

#258 Steinburg — 641 subs · avg R = 0.626 · H:169 C:110

#259 Wetteraukreis — 986 subs · avg R = 0.626 · H:263 C:157

#260 Greiz — 96 subs · avg R = 0.626 · H:28 C:15

#261 Hof, Landkreis — 126 subs · avg R = 0.626 · H:31 C:12

#262 Düsseldorf, Kreisfreie Stadt — 185 subs · avg R = 0.626 · H:46 C:30

#263 Waldshut — 252 subs · avg R = 0.626 · H:82 C:34

#264 Rhein-Pfalz-Kreis — 160 subs · avg R = 0.626 · H:45 C:24

#265 Rhein-Sieg-Kreis — 556 subs · avg R = 0.626 · H:156 C:84

#266 Ostallgäu — 136 subs · avg R = 0.626 · H:32 C:21

#267 Essen, Kreisfreie Stadt — 732 subs · avg R = 0.626 · H:214 C:96

#268 Worms, Kreisfreie Stadt — 33 subs · avg R = 0.626 · H:13 C:3

#269 Kiel, Kreisfreie Stadt — 187 subs · avg R = 0.626 · H:57 C:24

#270 Traunstein — 301 subs · avg R = 0.626 · H:83 C:47

#271 Leipzig — 348 subs · avg R = 0.626 · H:90 C:60

#272 Rhön-Grabfeld — 165 subs · avg R = 0.626 · H:42 C:28

#273 Region Hannover — 818 subs · avg R = 0.626 · H:223 C:129

#274 Hildburghausen — 102 subs · avg R = 0.626 · H:31 C:16

#275 Görlitz — 434 subs · avg R = 0.625 · H:128 C:66

#276 Marburg-Biedenkopf — 459 subs · avg R = 0.625 · H:119 C:61

#277 Bergstraße — 818 subs · avg R = 0.625 · H:229 C:120

#278 Odenwaldkreis — 434 subs · avg R = 0.625 · H:128 C:57

#279 Euskirchen — 423 subs · avg R = 0.625 · H:115 C:58

#280 Plön — 93 subs · avg R = 0.625 · H:26 C:11

#281 Coburg, Kreisfreie Stadt — 29 subs · avg R = 0.625 · H:7 C:6

#282 Sömmerda — 45 subs · avg R = 0.625 · H:10 C:8

#283 Olpe — 94 subs · avg R = 0.625 · H:27 C:18

#284 Lippe — 439 subs · avg R = 0.625 · H:124 C:71

#285 Enzkreis — 256 subs · avg R = 0.625 · H:75 C:34

#286 Märkischer Kreis — 434 subs · avg R = 0.625 · H:127 C:57

#287 Rostock, Kreisfreie Stadt — 435 subs · avg R = 0.625 · H:111 C:61

#288 Herzogtum Lauenburg — 131 subs · avg R = 0.625 · H:38 C:20

#289 München, Kreisfreie Stadt — 1384 subs · avg R = 0.625 · H:394 C:205

#290 Wiesbaden, Kreisfreie Stadt — 232 subs · avg R = 0.625 · H:67 C:29

#291 Emsland — 126 subs · avg R = 0.625 · H:34 C:17

#292 Tirschenreuth — 212 subs · avg R = 0.624 · H:48 C:29

#293 Kelheim — 198 subs · avg R = 0.624 · H:51 C:25

#294 Cuxhaven — 269 subs · avg R = 0.624 · H:83 C:35

#295 Kitzingen — 176 subs · avg R = 0.624 · H:37 C:26

#296 Dessau-Roßlau, Kreisfreie Stadt — 282 subs · avg R = 0.624 · H:74 C:46

#297 Prignitz — 226 subs · avg R = 0.624 · H:64 C:30

#298 Halle (Saale), Kreisfreie Stadt — 86 subs · avg R = 0.624 · H:28 C:14

#299 Kaiserslautern, Kreisfreie Stadt — 163 subs · avg R = 0.624 · H:39 C:27

#300 Waldeck-Frankenberg — 419 subs · avg R = 0.624 · H:107 C:66

#301 Anhalt-Bitterfeld — 148 subs · avg R = 0.624 · H:43 C:20

#302 Tuttlingen — 105 subs · avg R = 0.624 · H:28 C:16

#303 Sigmaringen — 154 subs · avg R = 0.624 · H:42 C:25

#304 Ostalbkreis — 156 subs · avg R = 0.624 · H:35 C:24

#305 Karlsruhe, Stadtkreis — 392 subs · avg R = 0.623 · H:103 C:58

#306 Erlangen-Höchstadt — 252 subs · avg R = 0.623 · H:71 C:36

#307 Rottweil — 158 subs · avg R = 0.623 · H:47 C:19

#308 Sonneberg — 95 subs · avg R = 0.623 · H:25 C:11

#309 Segeberg — 482 subs · avg R = 0.623 · H:113 C:67

| | | |
|---|---|---|
| #310 | Aschaffenburg, Landkreis | 217 subs · avg R = 0.623 · H:61 C:29 |
| #311 | Gießen, Landkreis | 890 subs · avg R = 0.623 · H:243 C:126 |
| #312 | Passau, Landkreis | 730 subs · avg R = 0.623 · H:190 C:104 |
| #313 | Potsdam-Mittelmark | 415 subs · avg R = 0.623 · H:109 C:58 |
| #314 | Bayreuth, Kreisfreie Stadt | 31 subs · avg R = 0.623 · H:5 C:9 |
| #315 | Gotha | 108 subs · avg R = 0.623 · H:31 C:13 |
| #316 | Steinfurt | 297 subs · avg R = 0.623 · H:83 C:43 |
| #317 | Hochtaunuskreis | 294 subs · avg R = 0.623 · H:93 C:36 |
| #318 | Herford | 477 subs · avg R = 0.623 · H:129 C:68 |
| #319 | Biberach | 242 subs · avg R = 0.623 · H:58 C:36 |
| #320 | Vechta | 123 subs · avg R = 0.623 · H:37 C:18 |
| #321 | Jerichower Land | 163 subs · avg R = 0.622 · H:43 C:23 |
| #322 | Lüchow-Dannenberg | 285 subs · avg R = 0.622 · H:69 C:48 |
| #323 | Rhein-Lahn-Kreis | 104 subs · avg R = 0.622 · H:29 C:16 |
| #324 | Kassel, Landkreis | 335 subs · avg R = 0.622 · H:92 C:52 |
| #325 | Roth | 374 subs · avg R = 0.622 · H:109 C:49 |
| #326 | Ludwigshafen am Rhein, Kreisfreie Stadt | 106 subs · avg R = 0.622 · H:39 C:6 |
| #327 | Suhl, Kreisfreie Stadt | 14 subs · avg R = 0.622 · H:2 C:3 |
| #328 | Trier-Saarburg | 127 subs · avg R = 0.622 · H:34 C:19 |
| #329 | Breisgau-Hochschwarzwald | 217 subs · avg R = 0.622 · H:69 C:31 |
| #330 | Bad Dürkheim | 130 subs · avg R = 0.622 · H:42 C:12 |
| #331 | Ansbach, Landkreis | 793 subs · avg R = 0.622 · H:208 C:119 |
| #332 | Landau in der Pfalz, Kreisfreie Stadt | 32 subs · avg R = 0.622 · H:9 C:3 |
| #333 | Recklinghausen | 493 subs · avg R = 0.622 · H:132 C:71 |
| #334 | Schwäbisch Hall | 186 subs · avg R = 0.622 · H:53 C:22 |
| #335 | Northeim | 311 subs · avg R = 0.622 · H:79 C:50 |
| #336 | Kassel, Kreisfreie Stadt | 168 subs · avg R = 0.622 · H:56 C:15 |
| #337 | Heidenheim | 137 subs · avg R = 0.621 · H:31 C:20 |
| #338 | Main-Tauber-Kreis | 199 subs · avg R = 0.621 · H:57 C:24 |
| #339 | Ebersberg | 369 subs · avg R = 0.621 · H:102 C:43 |

#340 Harz — 279 subs · avg R = 0.621 · H:84 C:33

#341 Fürth, Landkreis — 191 subs · avg R = 0.621 · H:51 C:28

#342 Neunkirchen — 169 subs · avg R = 0.621 · H:43 C:27

#343 Cloppenburg — 92 subs · avg R = 0.621 · H:25 C:17

#344 Lahn-Dill-Kreis — 507 subs · avg R = 0.621 · H:136 C:79

#345 Stendal — 182 subs · avg R = 0.621 · H:48 C:26

#346 Würzburg, Kreisfreie Stadt — 130 subs · avg R = 0.620 · H:32 C:19

#347 Forchheim — 258 subs · avg R = 0.620 · H:70 C:41

#348 Sächsische Schweiz-Osterzgebirge — 625 subs · avg R = 0.620 · H:143 C:93

#349 Ingolstadt, Kreisfreie Stadt — 199 subs · avg R = 0.620 · H:54 C:25

#350 Schleswig-Flensburg — 168 subs · avg R = 0.620 · H:36 C:23

#351 Pforzheim, Stadtkreis — 124 subs · avg R = 0.620 · H:38 C:12

#352 Heidelberg, Stadtkreis — 167 subs · avg R = 0.620 · H:47 C:27

#353 Borken — 129 subs · avg R = 0.620 · H:32 C:26

#354 Harburg — 207 subs · avg R = 0.620 · H:48 C:32

#355 Darmstadt, Kreisfreie Stadt — 274 subs · avg R = 0.620 · H:77 C:33

#356 Neuwied — 250 subs · avg R = 0.619 · H:67 C:27

#357 Haßberge — 165 subs · avg R = 0.619 · H:45 C:28

#358 Börde — 115 subs · avg R = 0.619 · H:28 C:16

#359 Bad Tölz-Wolfratshausen — 189 subs · avg R = 0.619 · H:51 C:26

#360 Stade — 74 subs · avg R = 0.619 · H:14 C:8

#361 Saarpfalz-Kreis — 179 subs · avg R = 0.619 · H:51 C:28

#362 Nienburg (Weser) — 103 subs · avg R = 0.619 · H:25 C:14

#363 Helmstedt — 72 subs · avg R = 0.619 · H:18 C:13

#364 Verden — 100 subs · avg R = 0.618 · H:28 C:12

#365 Weimar, Kreisfreie Stadt — 49 subs · avg R = 0.618 · H:12 C:12

#366 Erding — 367 subs · avg R = 0.618 · H:110 C:48

#367 Gelsenkirchen, Kreisfreie Stadt — 119 subs · avg R = 0.618 · H:36 C:16

#368 Lichtenfels — 120 subs · avg R = 0.617 · H:38 C:15

#369 Neustadt an der Weinstraße, Kreisfreie Stadt — 39 subs · avg R = 0.617 · H:9 C:7

#370 Heilbronn, Stadtkreis 55 subs · avg R = 0.617 · H:14 C:9

#371 Braunschweig, Kreisfreie Stadt 154 subs · avg R = 0.617 · H:45 C:22

#372 Kyffhäuserkreis 120 subs · avg R = 0.617 · H:30 C:21

#373 Kulmbach 68 subs · avg R = 0.617 · H:19 C:10

#374 Ilm-Kreis 219 subs · avg R = 0.616 · H:61 C:29

#375 Zollernalbkreis 117 subs · avg R = 0.616 · H:35 C:14

#376 Heidekreis 87 subs · avg R = 0.616 · H:28 C:6

#377 Schaumburg 289 subs · avg R = 0.616 · H:82 C:33

#378 Würzburg, Landkreis 205 subs · avg R = 0.616 · H:55 C:31

#379 Landkreis Rostock 305 subs · avg R = 0.616 · H:82 C:30

#380 Saale-Orla-Kreis 118 subs · avg R = 0.616 · H:26 C:19

#381 Kronach 97 subs · avg R = 0.615 · H:24 C:12

#382 Leverkusen, Kreisfreie Stadt 26 subs · avg R = 0.615 · H:7 C:4

#383 Kaufbeuren, Kreisfreie Stadt 7 subs · avg R = 0.615 · H:1

#384 Leer 134 subs · avg R = 0.615 · H:38 C:13

#385 Neuburg-Schrobenhausen 48 subs · avg R = 0.614 · H:10 C:10

#386 Koblenz, Kreisfreie Stadt 50 subs · avg R = 0.614 · H:10 C:10

#387 Burgenlandkreis 152 subs · avg R = 0.613 · H:41 C:14

#388 Osnabrück, Landkreis 171 subs · avg R = 0.613 · H:37 C:27

#389 Krefeld, Kreisfreie Stadt 72 subs · avg R = 0.612 · H:22 C:5

#390 Pfaffenhofen a. d. Ilm 136 subs · avg R = 0.611 · H:27 C:21

#391 Mansfeld-Südharz 166 subs · avg R = 0.610 · H:40 C:25

#392 Wesermarsch 61 subs · avg R = 0.610 · H:11 C:9

#393 Mainz, Kreisfreie Stadt 43 subs · avg R = 0.610 · H:8 C:5

#394 Coburg, Landkreis 73 subs · avg R = 0.610 · H:18 C:9

#395 Oberhausen, Kreisfreie Stadt 31 subs · avg R = 0.610 · H:9 C:5

#396 Dillingen a.d. Donau 61 subs · avg R = 0.608 · H:17 C:7

#397 Unna 125 subs · avg R = 0.608 · H:39 C:14

#398 Friesland (DE) 45 subs · avg R = 0.606 · H:7 C:8

#399 Eichsfeld 107 subs · avg R = 0.605 · H:24 C:14

#400 Regensburg, Kreisfreie Stadt — 88 subs · avg R = 0.604 · H:20 C:12

#401 Wittmund — 36 subs · avg R = 0.603 · H:11 C:2

#402 Saale-Holzland-Kreis — 59 subs · avg R = 0.602 · H:14 C:5

#403 Zweibrücken, Kreisfreie Stadt — 40 subs · avg R = 0.598 · H:14 C:1

#404 Hof, Kreisfreie Stadt — 25 subs · avg R = 0.597 · H:4

#405 Brandenburg an der Havel, Kreisfreie Stadt — 53 subs · avg R = 0.597 · H:12 C:4

#406 Memmingen, Kreisfreie Stadt — 5 subs · avg R = 0.595 · H:2

#407 Nordrhein-Westfalen — 1 subs · avg R = 0.594 · H:0

#408 Bayern — 6 subs · avg R = 0.582 · H:1

#409 Amberg, Kreisfreie Stadt — 23 subs · avg R = 0.578 · H:5 C:1

The Saarland ranks **#1 of 409 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Saarland, Schleswig-Holstein, Sachsen, while the three most resilient are Amberg, Kreisfreie Stadt, Bayern, Nordrhein-Westfalen. 201 of 409 score above the fleet average of 0.628. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

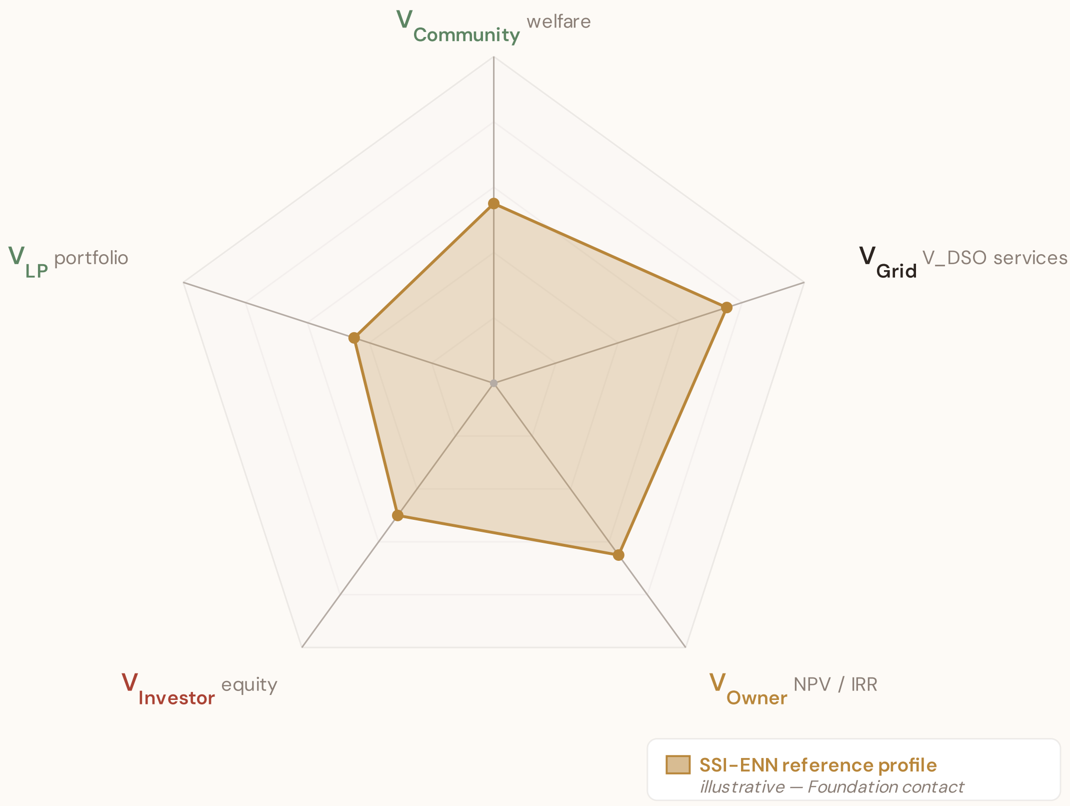
SECTION F — RECURRING MODULE

## SSI Enhanced Neural Network — Monthly Topic

**From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

### SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): $V_{Community}$ (welfare avoided outages, energy-poverty relief, employment) · $V_{Grid}$ (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · $V_{Owner}$ (NPV / IRR / payback per asset class) · $V_{Investor}$ (risk-adjusted equity return at the financing tier) · $V_{LP}$ (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.




*Editorial reference shape — $V_{Grid}$ dominates (V_DSO captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community}$ + $V_{Owner}$ are co-secondary. $V_{Investor}$ + $V_{LP}$ reflect the financing-tier waterfall above $V_{Owner}$ NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.*

| Layer 1 — Data | Layer 2 — Valuation | Layer 3 — Neural Net | Layer 4 — Coverage | Layer 5 — Platform |
|---|---|---|---|---|
| SSI v4.0 component taxonomy, normalisation, MC engine | DSO/TSO + Community value stacks, TCO, real options | GBT+DNN ensemble, transfer learning, SHAP | Country prioritisation, DCI, registry, calibration | Products, API, revenue model, unit economics |

**LAYER 1** §0.3.4 · September 2026

## Gaussian Copula Monte Carlo Engine

*How 10,000 simulations per substation capture the correlation structure across grid risk factors*

The SSI components are not independent — a substation with poor continuity (high C) often also has poor voltage quality (high V), aged infrastructure (high I5), and is located in a rural area with high economic vulnerability (high E). The Gaussian copula captures these dependencies through a 20×20 correlation matrix estimated from the national fleet, decomposed via Cholesky factorisation to generate correlated random draws. For each substation, 10,000 Monte Carlo iterations propagate uncertainty through the full scoring formula, producing not just a point estimate (R_median) but a complete posterior distribution — P5, P50, P95, confidence interval width, and skewness. This is the computational backbone, yielding the uncertainty quantification that distinguishes the SSI from deterministic indices.

KEY CONCEPTS

- → 20×20 Gaussian copula with Cholesky decomposition
- → 10,000 iterations per substation
- → Full posterior: R_P5, R_median, R_P95, CI_width, skewness
- → Correlation captures systemic clustering of risk factors

LIVE SSI DATA CONNECTION

The SSI Index currently scores **108,016 substations**. Fleet mean R = 0.628, with an average confidence interval width of 0.220 from 10,000 MC iterations per substation. The dominant component is C (Continuity) at avg 0.350 / 0.30 ceiling (117% utilisation).

**Coming next month:** LAYER 5 Platform — Product Delivery Pipeline — *From raw analytical outputs to four commercial product formats*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

## Radar 2 — W1–W10 Substation Community Value

**Wall 1 — Public W1–W10 framework.** The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

### 5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

**Published (5)** — solid fill
W1 · W2 · W3 · W4 · W8
*Peer-reviewed methodology operational at fleet scale.*

**Published-baseline (1) ▲** — solid fill + chevron corner marker
W9 — Supply Security
*Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).*

**Baseline-only (1)** — striped fill
W5 — Digital Inclusion
*Foundation methodology under development; baseline value defensible at v4.2.*

**Commercial (3)** — dashed-faded
W6 · W7 · W10
*Commercial deliverables — please contact the Foundation.*

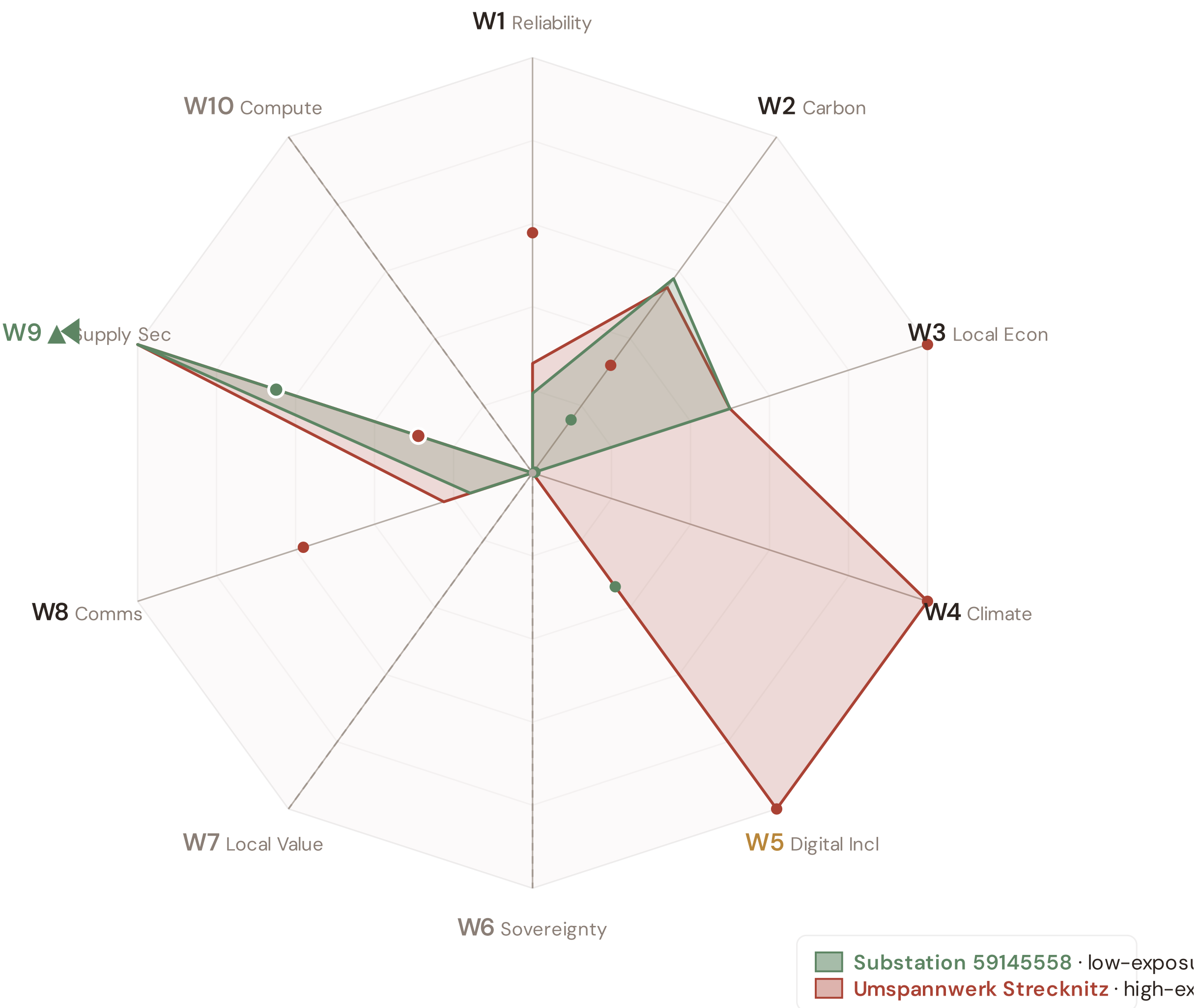
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

### v4.2 methodology disclosure — per-country normalisation

Across Germany, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to **[0, 1]** via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (`SSI_v4_0 Italy/.../scoring-it/normalise.py`); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10-d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

## Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **[DE_59145558, 110.0 kV, Segeberg]** (low-exposure reference, Re_norm 0.000) and **Helmstorf** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.




*Substation 59145558 polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Umspannwerk Strecknitz polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.*

## W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE_W_BASELINES.PY

| AXIS | NAME | AUDIT STATE | METHODOLOGY BASIS |
|---|---|---|---|
| **W1** | Reliability | **Published** | C × E2_local × population ( SAIDI + + -POV) |
| **W2** | Carbon | **Published** | T1 × zonal MEF × (V · deg / 100) ( + EEA) |
| **W3** | Local Economic | **Published** | norm(R10_just) — EU-SILC + OIPE energy-poverty |
| **W4** | Climate Resilience | **Published** | (R6d − 1) + (R6e − 1) + (R6c − 1) + ½(R9 − 1) — v4.2 resilience modifier family |
| **W5** | Digital Inclusion | **Baseline-only** | norm(R8_adapt) — bulk SM + + DESI |
| **W6** | Sovereignty | **Commercial** | Foundation contact (ssi_index@ikenga.eu) |
| **W7** | Local Value Retention | **Commercial** | Foundation contact (ssi_index@ikenga.eu) |
| **W8** | Communications | **Published** | C × p_crit_10yr × population (Markov 10-yr critical probability) |
| **W9 ▲** | Supply Security | **Published-baseline ▲** | 0.60 × BC_pct + 0.30 × bridge + 0.10 × (1 − deg / 20) — OSM power-graph betweenness centrality + bridge identification + degree-based redundancy |
| **W10** | Compute Multiplier | **Commercial** | Foundation contact (ssi_index@ikenga.eu) |

### ▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

**Worked examples.** The W1-W10 per-substation baseline values for two anchor cases — Substation 59145558 (low-exposure reference) and Umspannwerk Strecknitz (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max Re_norm) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

**Anti-maladaptation discipline.** The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

## Looking Ahead

NEXT MONTH — EDITION 05

### Alpine Valleys

Deep-dive into Region 05's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Narrow valleys, long restoration MTTR.

NEXT DATA REFRESH

### Q4 2026 — 0 Quarterly Sources

**none** are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **BNetzA (Bundesnetzagentur)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

### Seismic Zone 4–5: 23815 Substations in Highest Hazard

**23815 substations** sit in the highest seismic hazard zones (Zones 4–5, moderate to strong). Of these, **11812** already score High/Critical — meaning their grid condition is poor *before* considering earthquake risk. The R6b seismic modifier captures this compound vulnerability.

Edition 05 will be published on the second Thursday of October 2026. Deep-dive region: **Region 05**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

**Feedback welcome.** The SSI Index is designed to evolve through stakeholder dialogue. If you represent an German utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 006

Published 10 September 2026 · SSI Index **v4.2** (refreshed June 2026) · 108,016 substations · 262,134 grid lines · 6 components + **Re composite** (Radar 1 7th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

### Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.